

FORECASTING REAL ESTATE RENTAL PRICES

Group 22

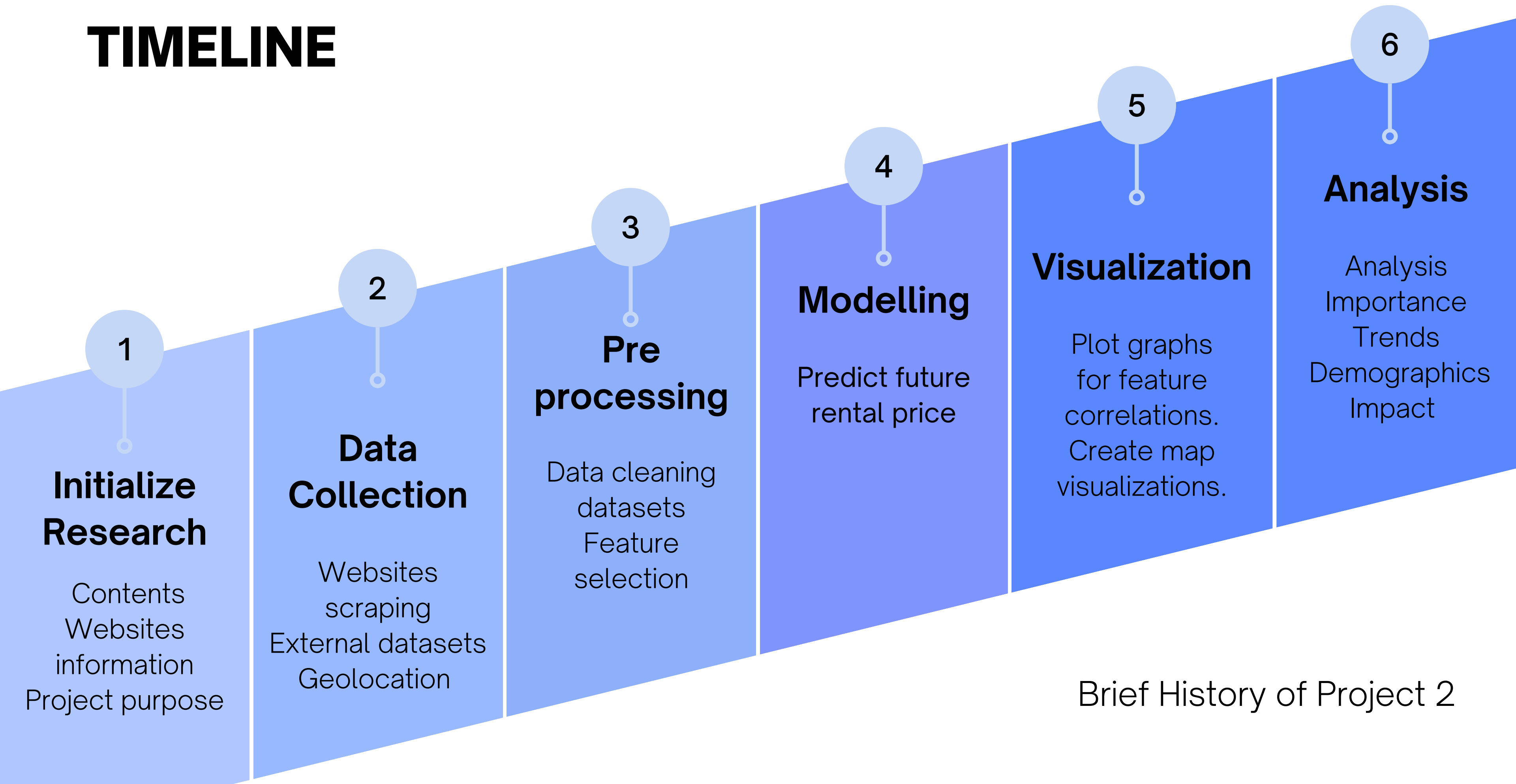
- Zihan Yin
- Ximin Wan
- Xiaoqing Lei
- Suhui Tu



A collage of logos and illustrations representing data science and finance. It includes the pandas logo, Apache Spark logo, a Python logo, a blue house, a balance scale with dollar signs, a magnifying glass, a calculator, and a person. The background is a light blue circle.

- **Most important features in predicting rental prices**
- **The top 10 suburbs with the highest predicted growth rate**
- **The most liveable and affordable suburbs according to your chosen metrics**

TIMELINE



Brief History of Project 2

Visualization
15%

Data Collection
10%

WORKHOURS
150H

Modelling
15%

Preprocessing
60%

Group meeting
20-25h/wk

Communication Tools



Files sharing



DATA INTRODUCTION





DATA INTRODUCTION

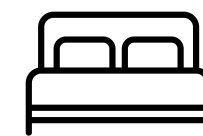
Features in Main dataset of property in
VIC, Australia from Domain.com



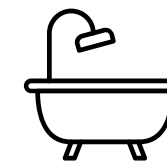
Property Name



Property Coordinate



Number of Bedrooms



Number of Bath Rooms



Number of Parkings

DATA INTRODUCTION

SA2 info (From ABS)

- SA2 District Code
- SA2 District Name
- Geometry Boundary of SA2 District

External Dataset

- Personal Income
(Fm ABS)



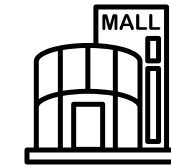
- Population Density
(From ABS)



- Offence Count
(From vic.gov)



Spatial Proximity Information



Closest **Mall** Distance
(Fm: [australia shoppings.com](https://australia.shoppings.com))



Closest **Hospital** Distance
(Fm: vic.gov)



Closest **School** Distance
(Fm: vic.gov)



Closest **Station** Distance
(From vic.gov)



Closest **Parks** Distance
(From vic.gov)



Closest **CBD** Distance

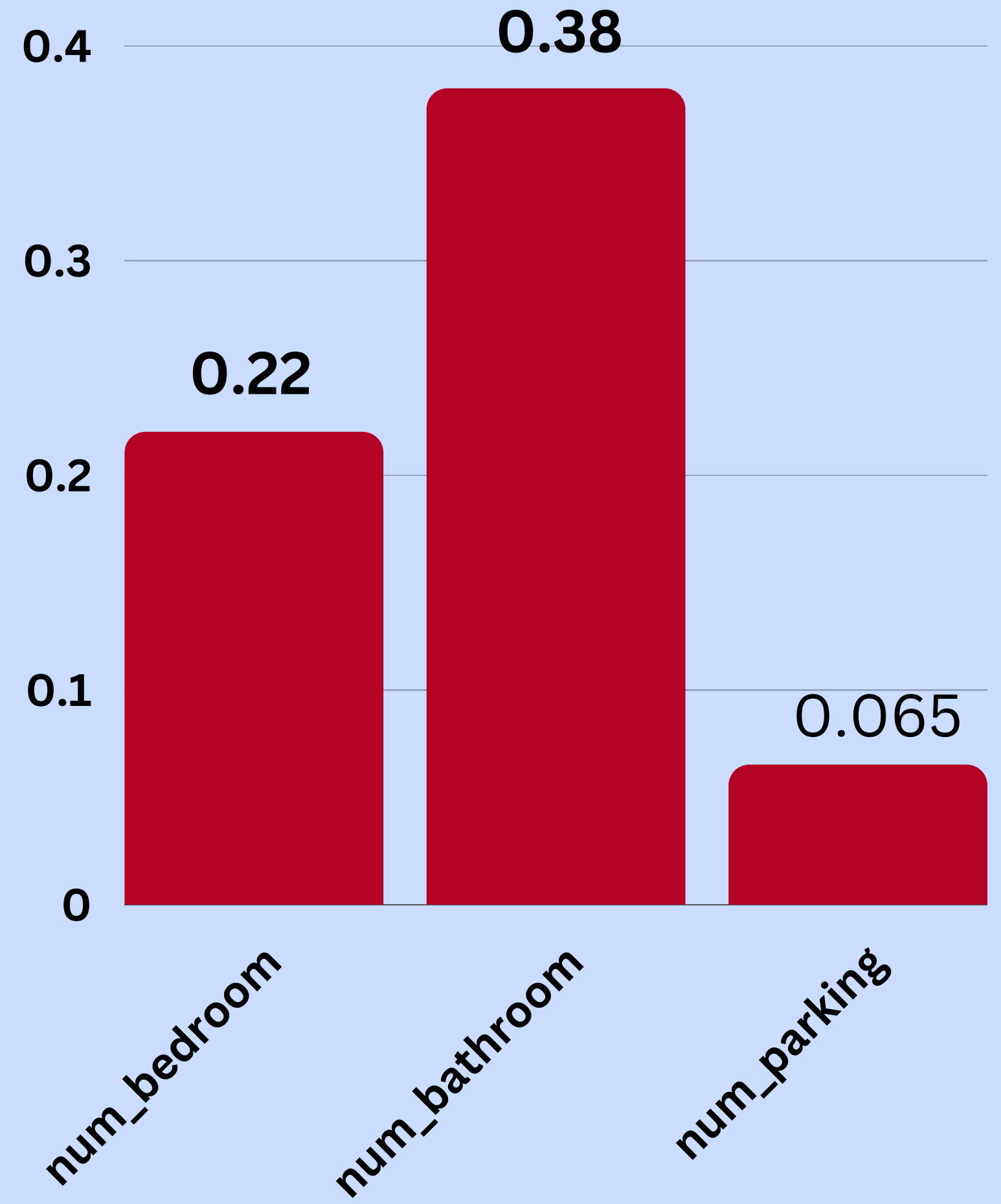
THE MOST IMPORTANT FEATURES IN PREDICTING RENTAL PRICES



MOST IMPORTANT FEATURES

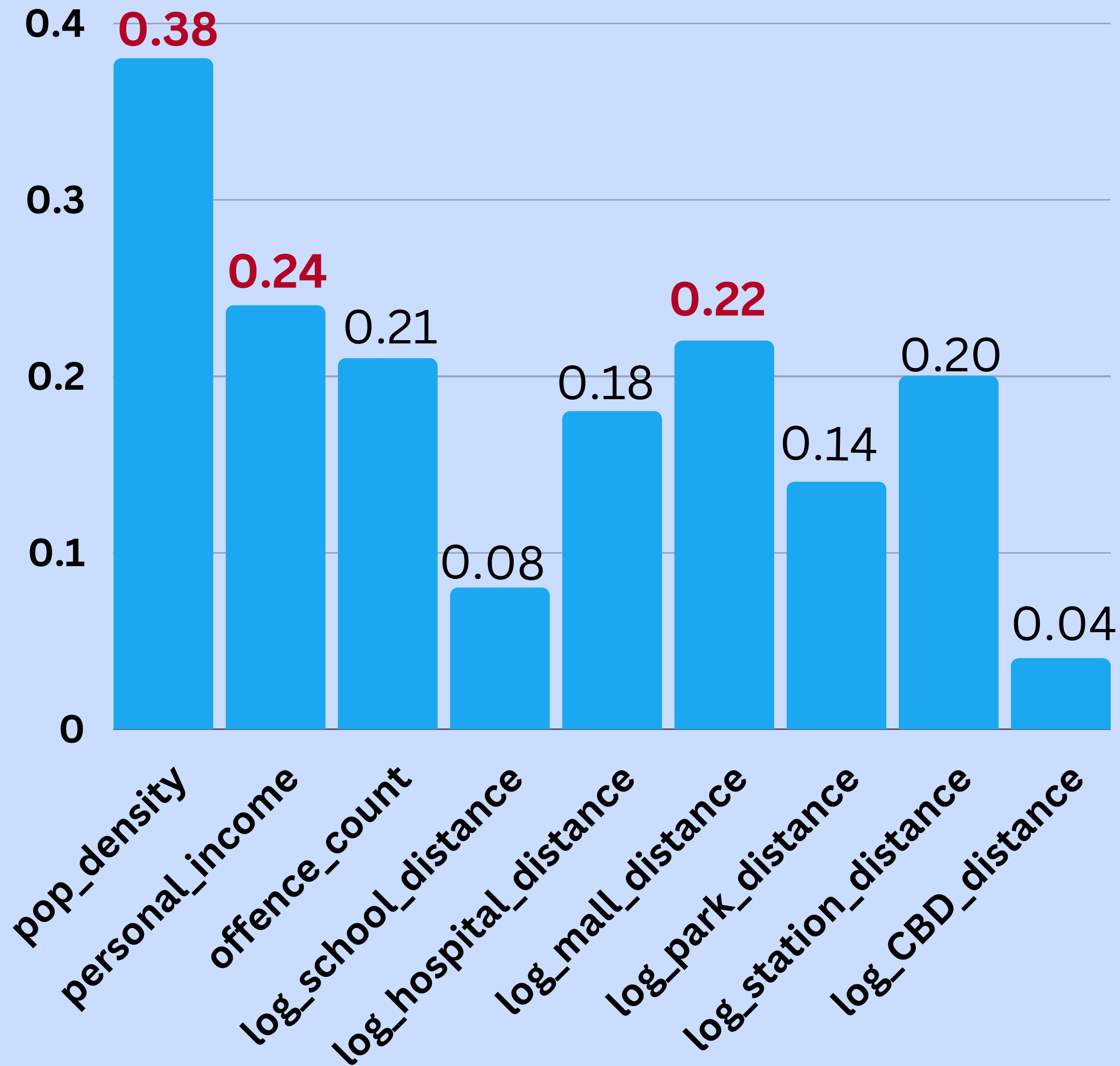
Spearman Correlation- Internal

Number of parking is not an important feature which affect the rental price.

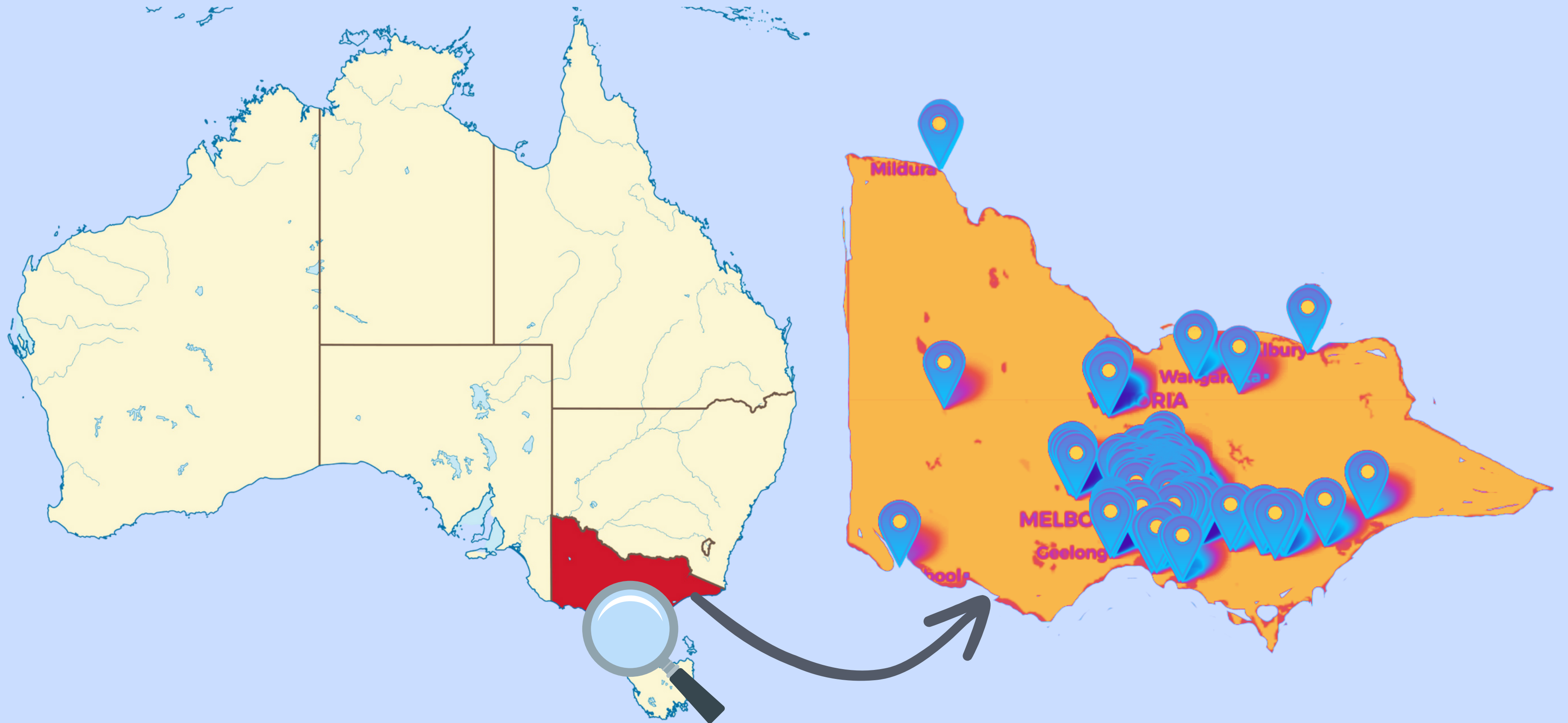


Spearman Correlation -External

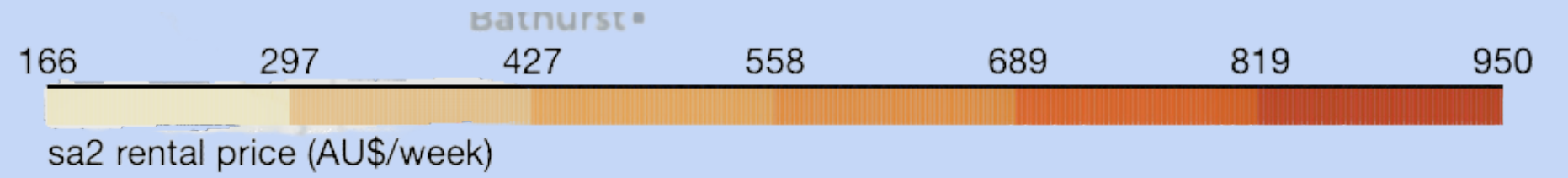
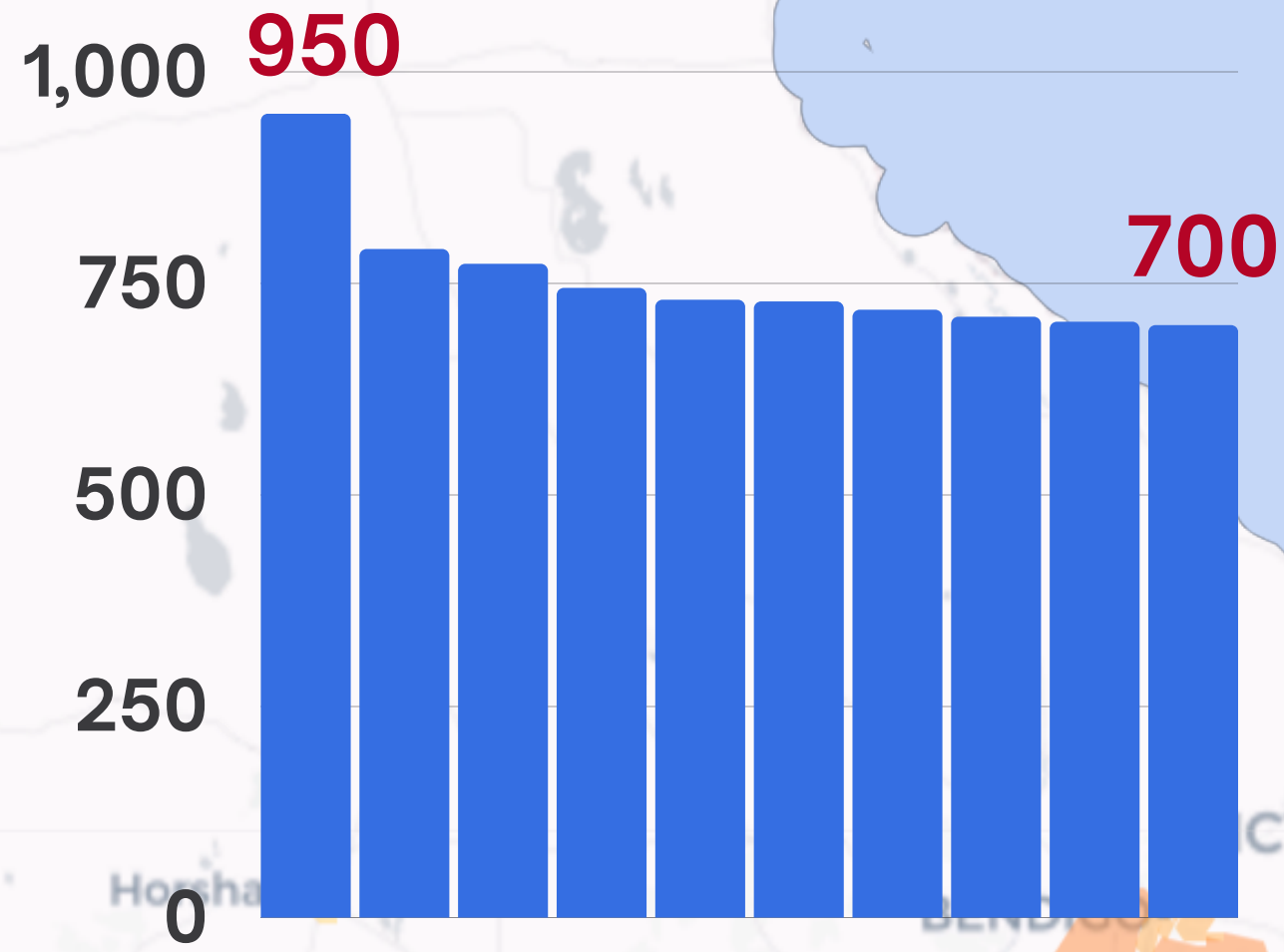
- Population density and personal income has a weak positive correlation.
- Offence count, closest mall distance and closest station distance have weak negative correlation.



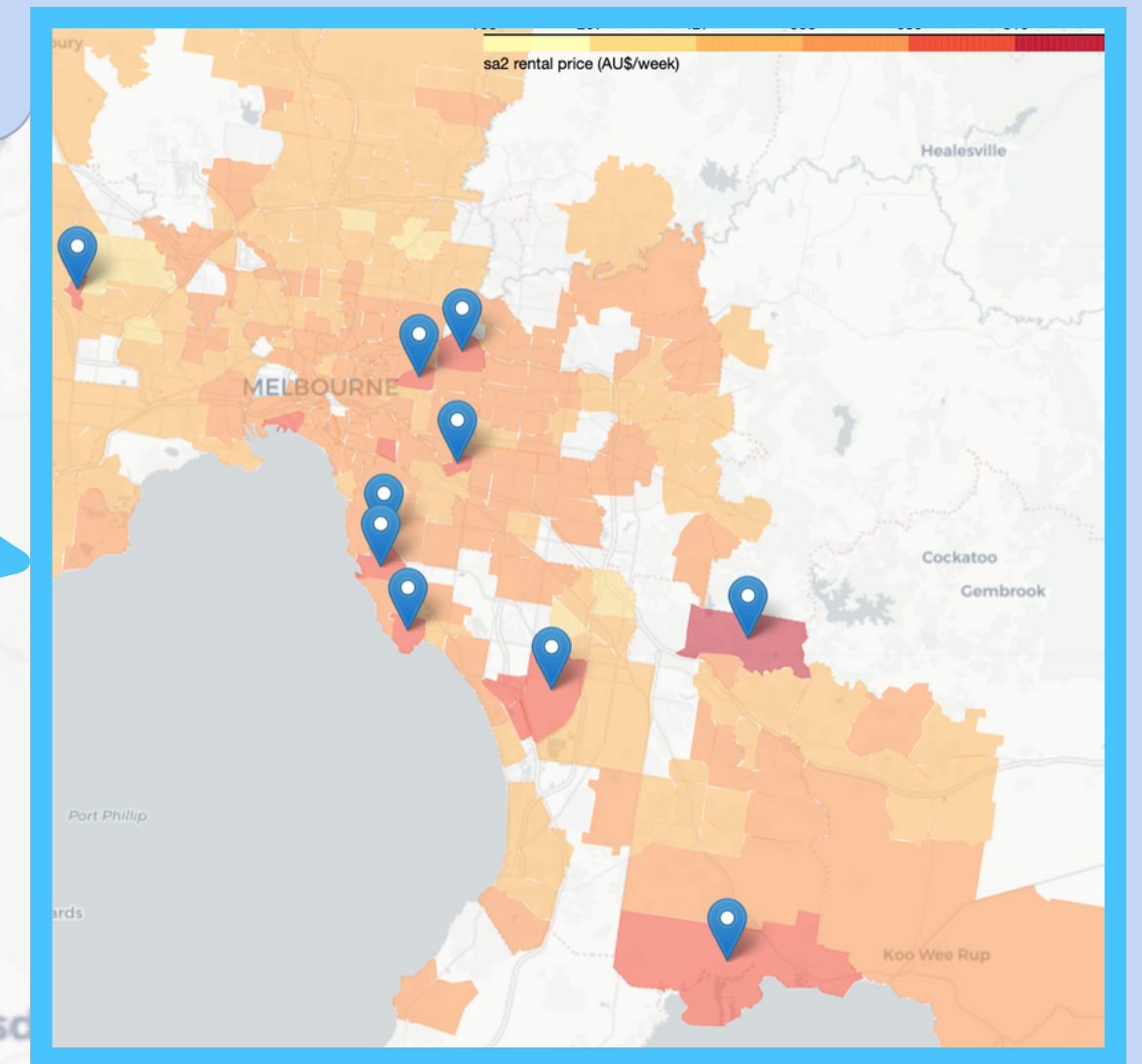
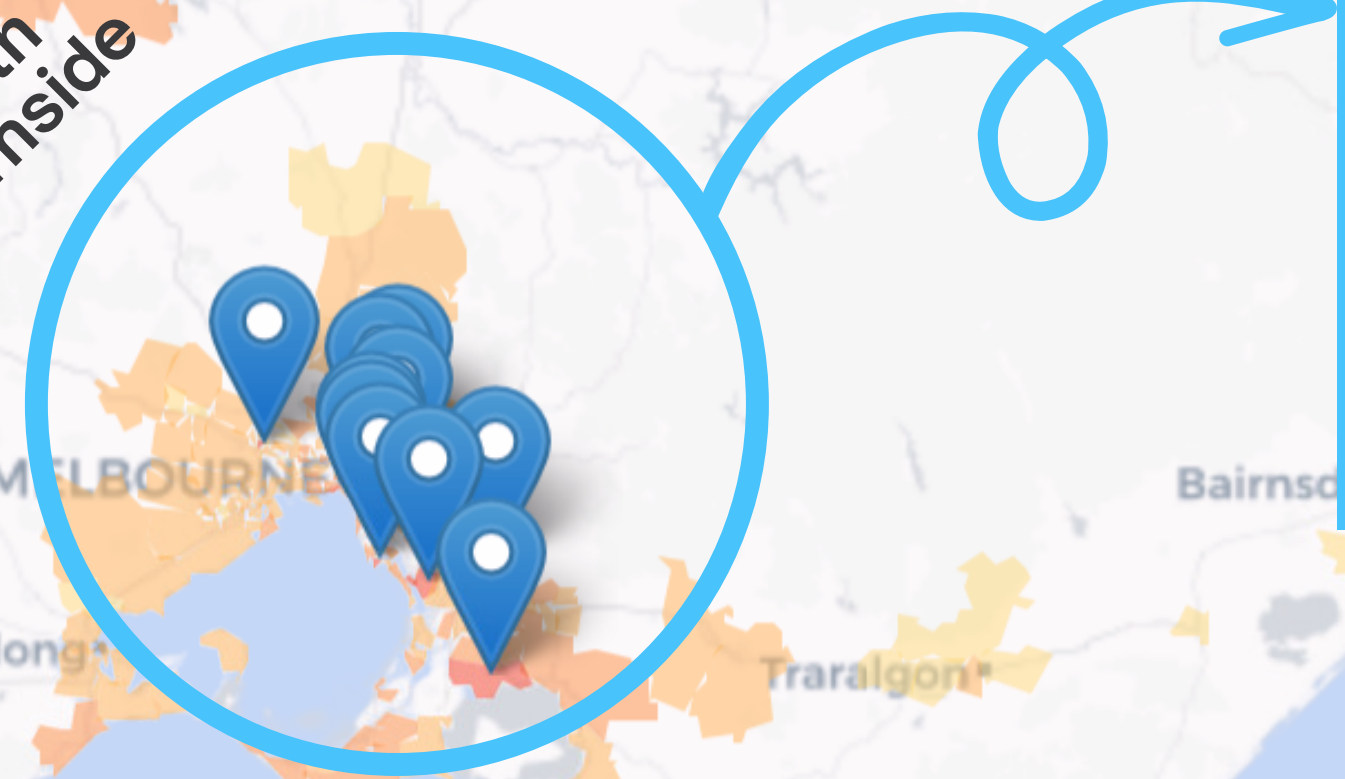
Distribution of Sampling Property



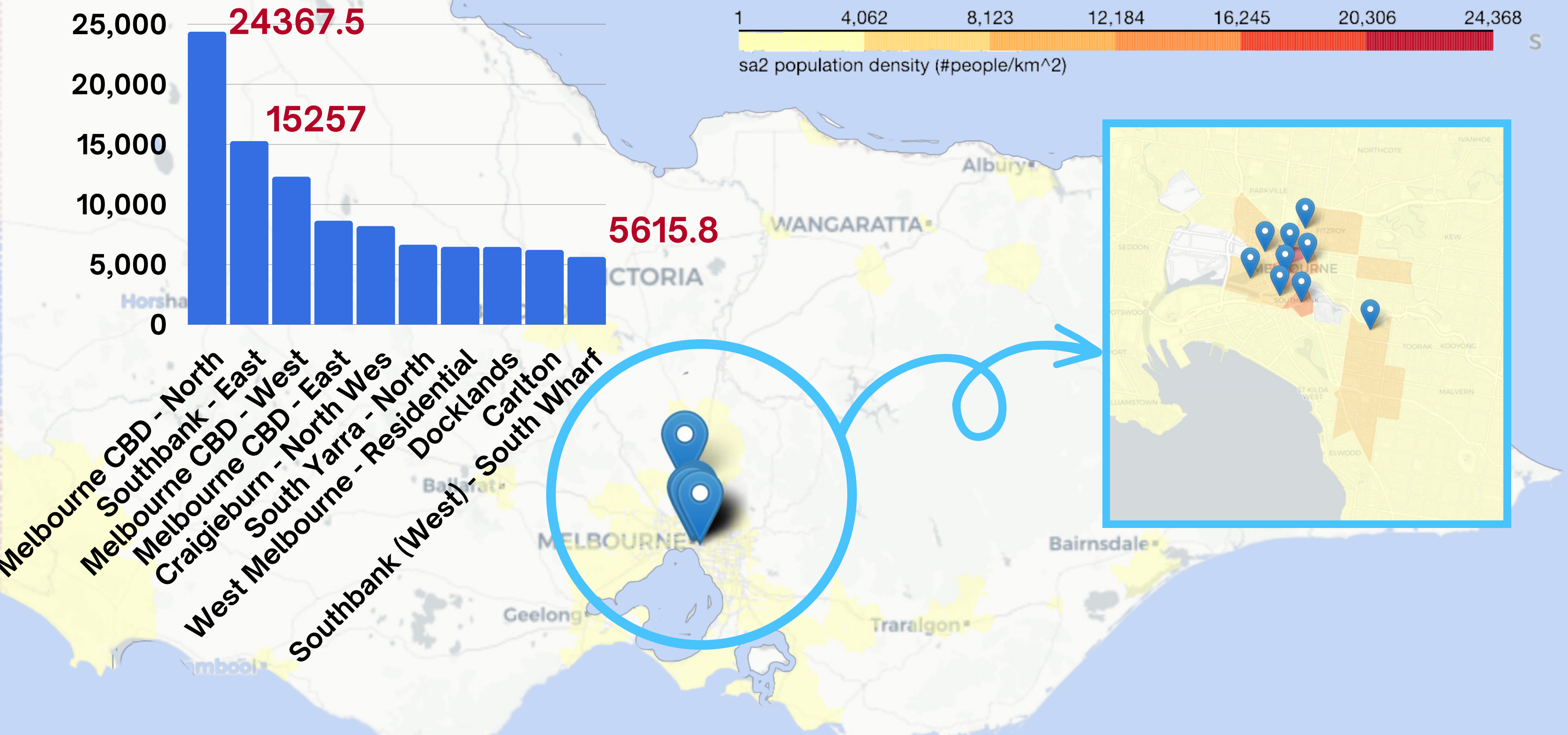
Top 10 Average Rental Price (\$) of SA2 District



Narre Warren North
Beaumaris
Brighton East
Ashburton(Vic)
Hampton
Pearcedale-Tooradin
Kew-South
Keysborough-South
Balwyn North
Burnside



Top 10 Population Density(people/km²) by SA2 District



TOP 10 SUBURBS WITH HIGHEST GROWTH RATE



GENERATE 2026 DATA

2023

`population density`
`personal income`
`offence count`

**rate of
change**

2026

`Population Density`
`Personal Income`
`Offence Count`

2023

Other features

Same

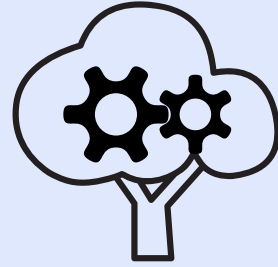
2026

Other features



Models

01 - Random Forest Regression



Usually produces relatively **stable predictions**.

Avoid problems caused by **outliers** and **noise** in data.

02 - Neural Network - NNW



Mimics human **brain processes** information.

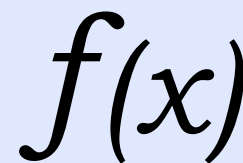
Do well in **dealing complex non-linear** relationships.

03 - K-Nearest Neighbors (KNN)



Properties with **similar conditions** and features might be **priced nearly**.

04 - General Linear Model (GLM)



Uses **different families** of distributions to establish **non-linear** relationships.

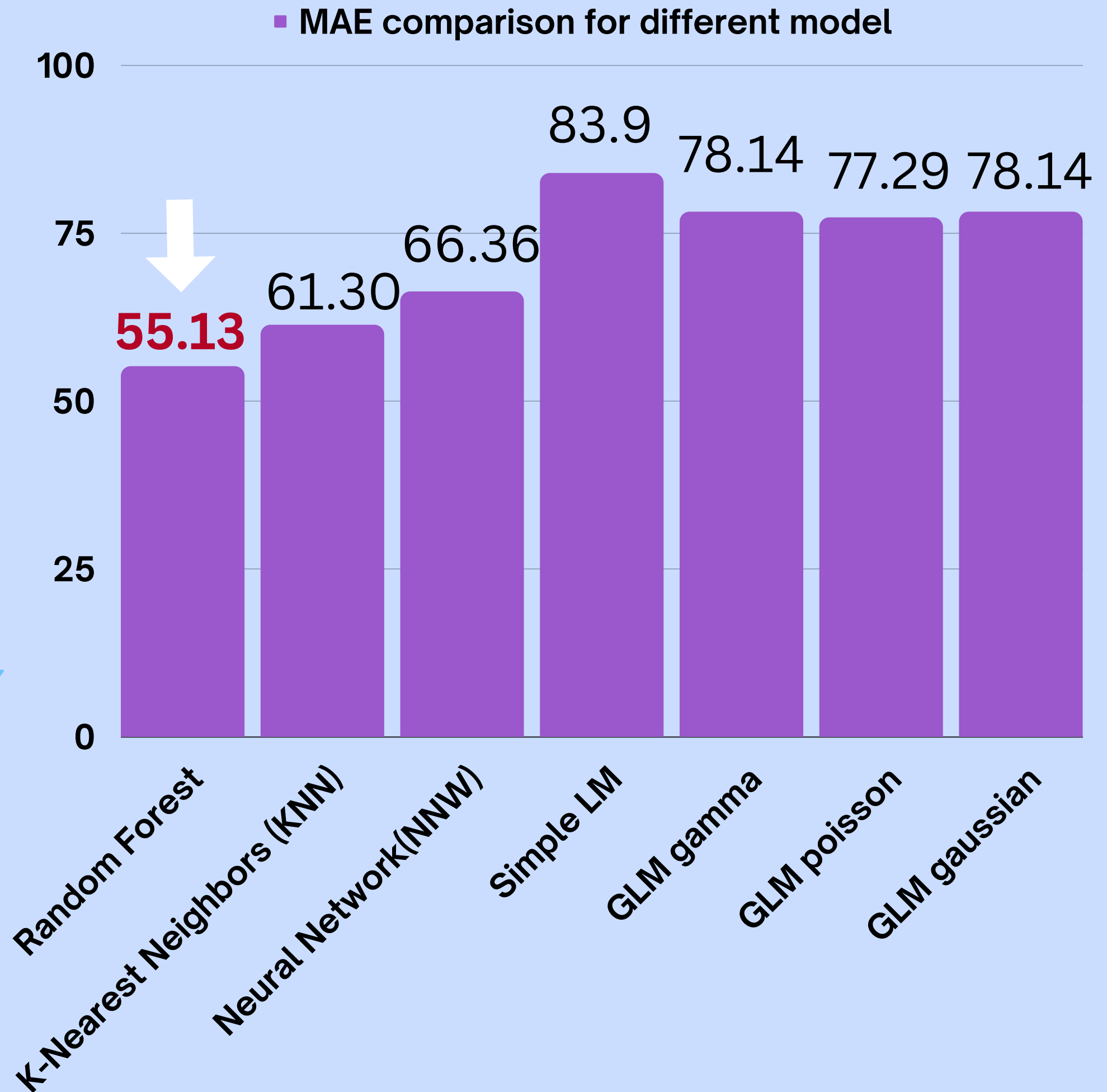
More **directly show** the **relationship** between house prices and different features.

MODEL'S PERFORMANCE

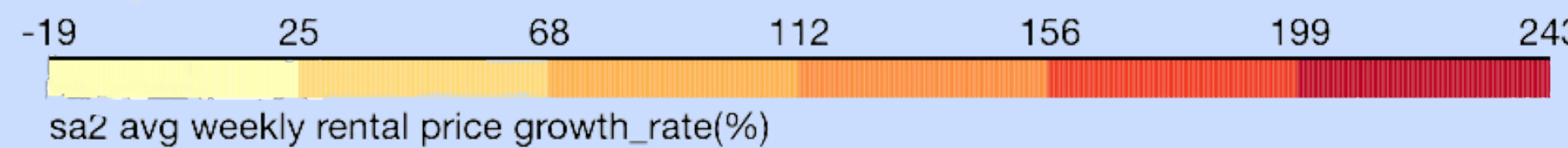
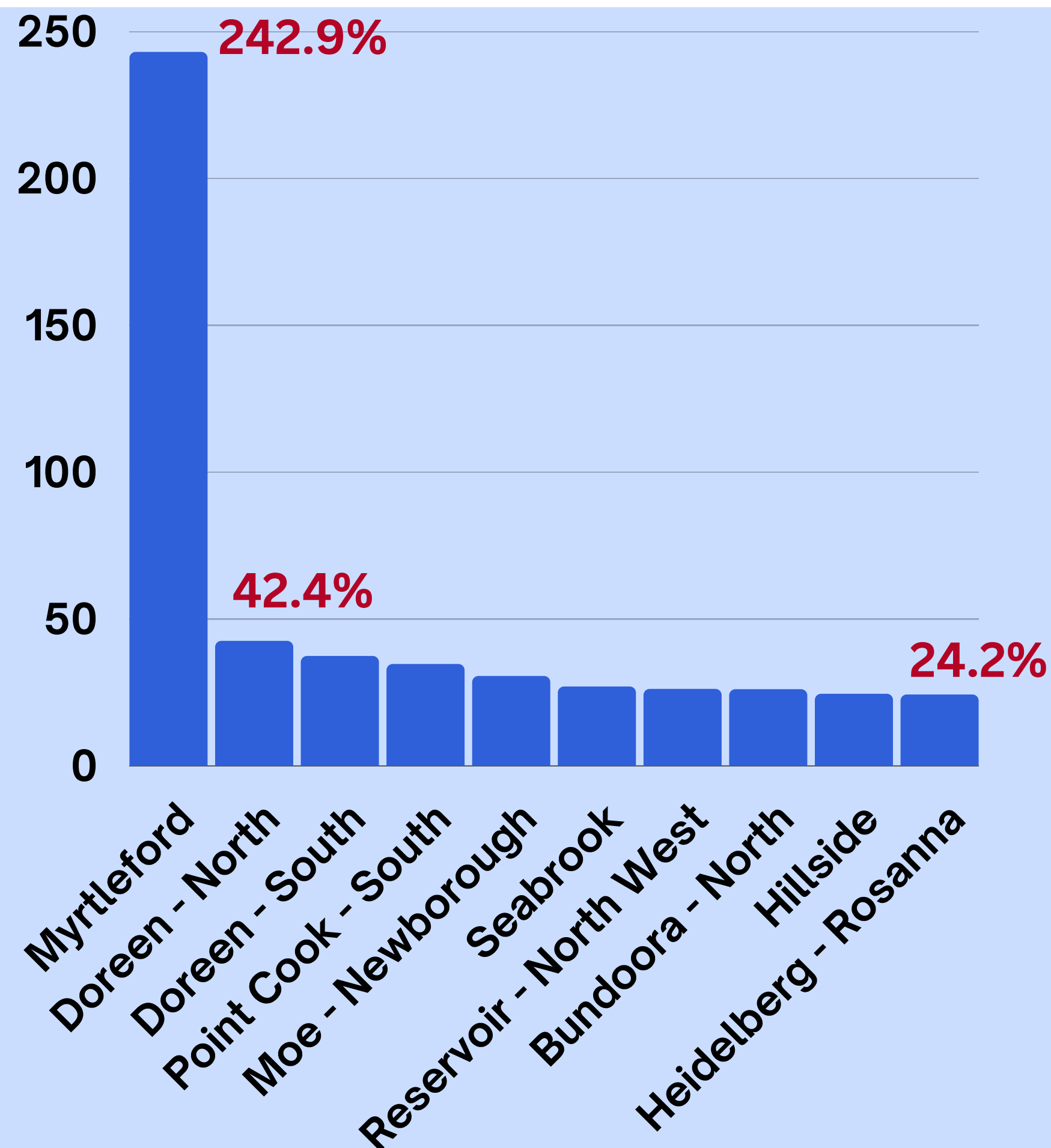
Random forest model has the lowest mean absolute error (MAE) around 55 (**best**).

This model will be used to predict the rental price for each property in 2026.

**Lower MAE,
Better Model Performance.**



Top 10 - SA2 rental price growth rates (2023-2026)

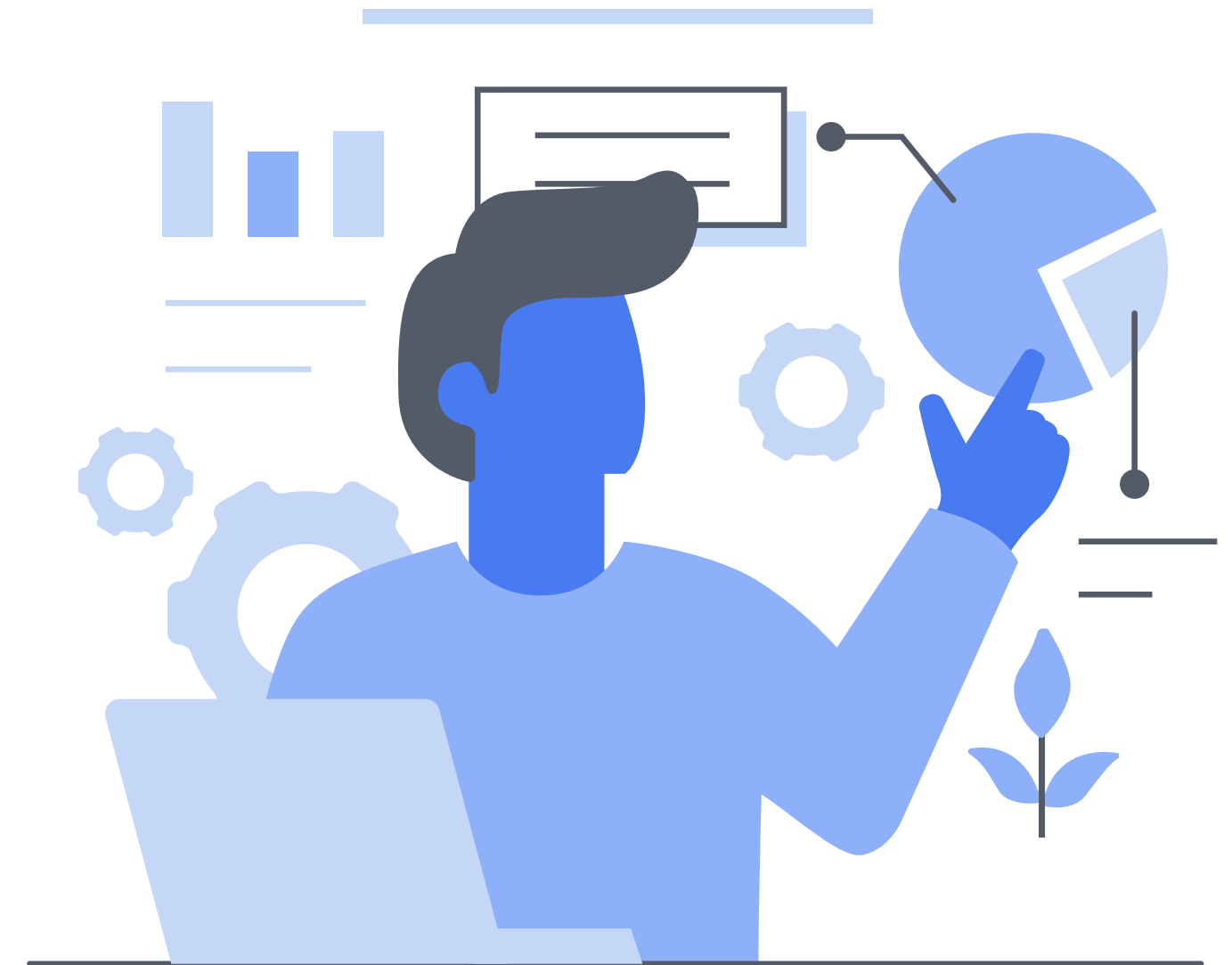


THE MOST LIVEABLE AND AFFORDABLE SUBURBS



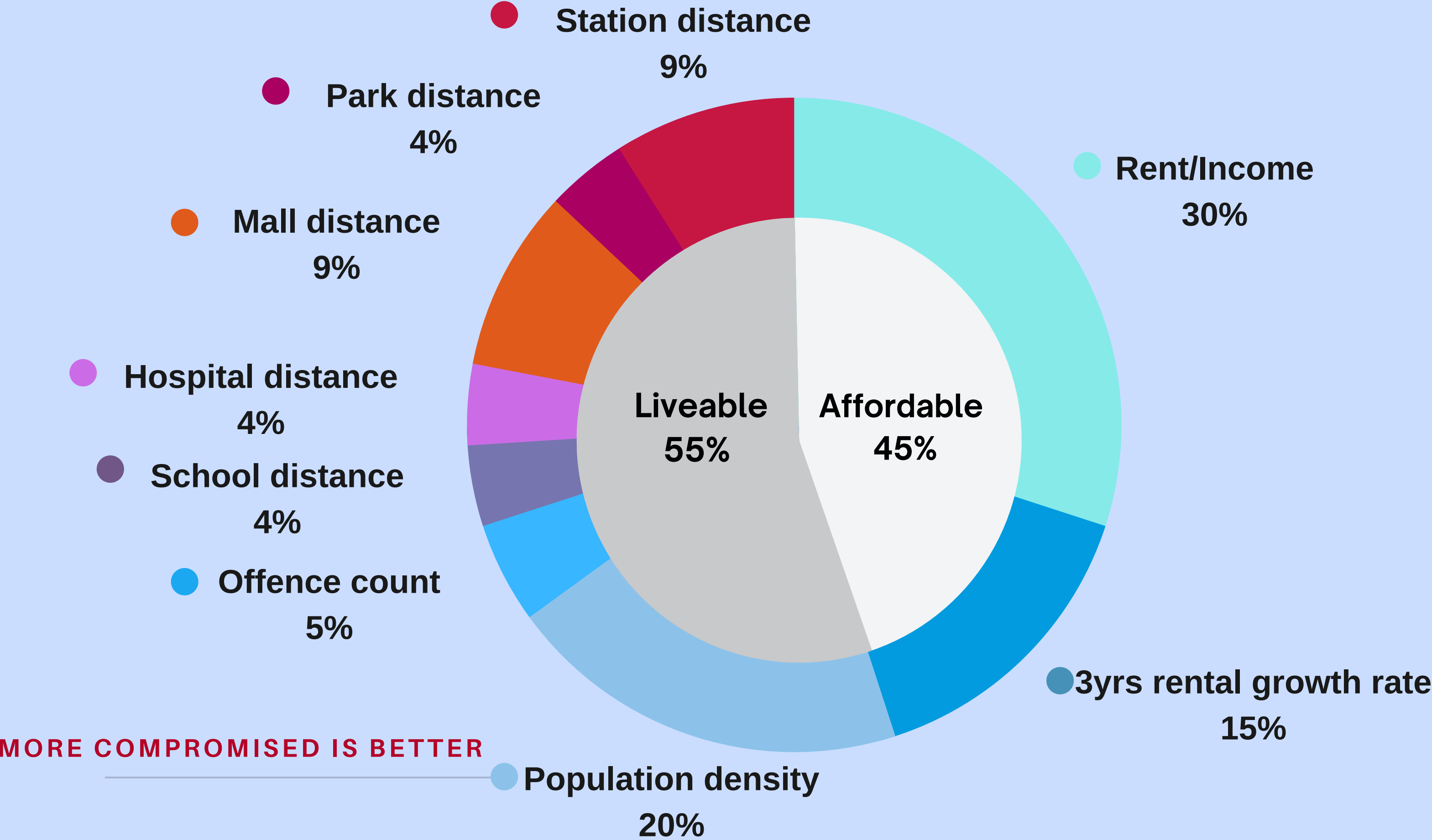
Based on Our Experience:

- We first consider whether we can afford to rent this property now and in the future (45%).
- On this basis, we will consider the livability of the suburb (55%).
- Then...



**SA2 SCORING
SYSTEM**

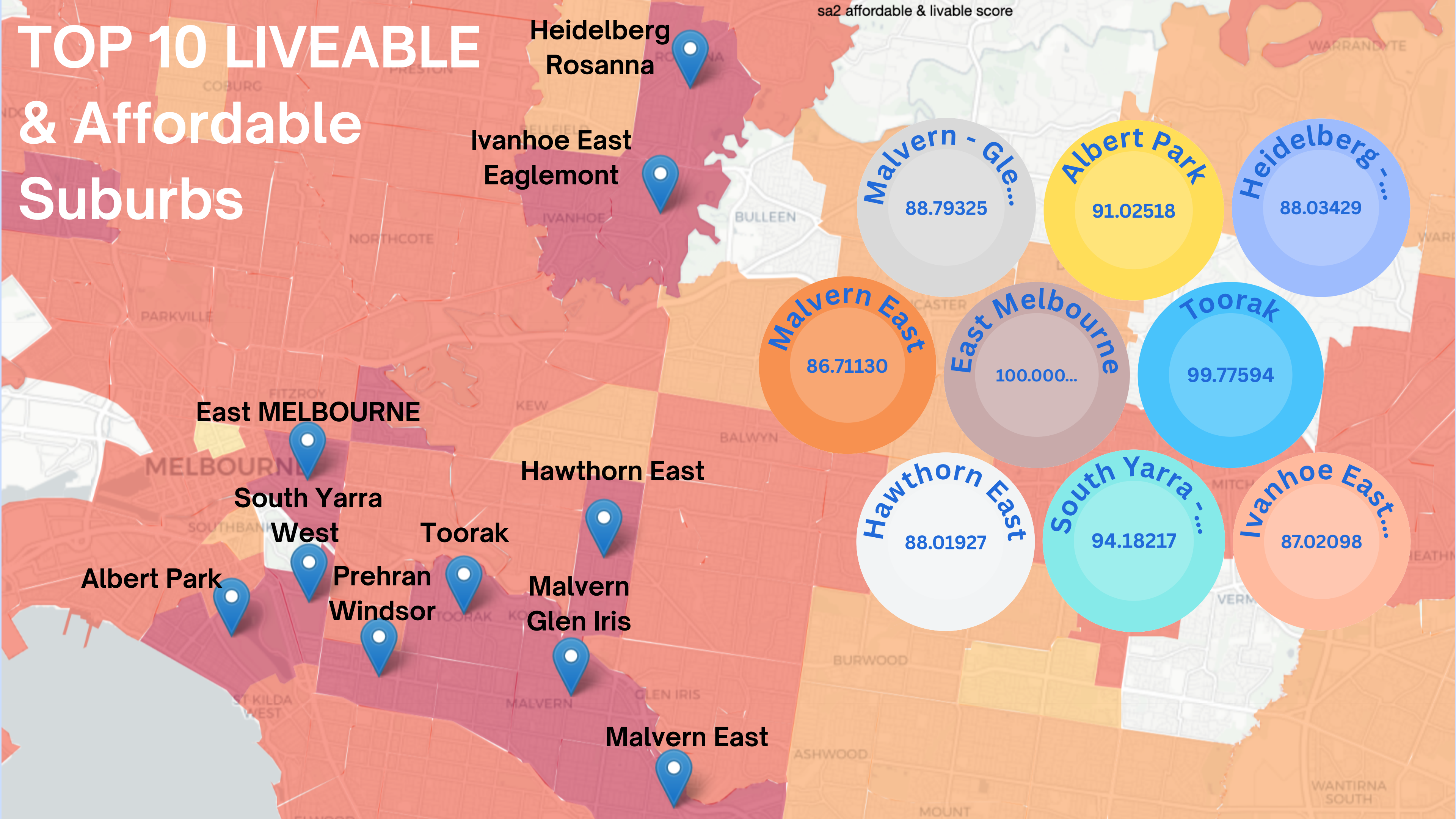
Livable & Affordable Score



Top 10 Affordable & Liveable Suburbs



SA2 Name	Affordable & Liveable Score
East Melbourne	100
Toorak	99.78
South Yarra - West	94.18
Albert Park	91.03
Malvern - Glen Iris	88.79
Heidelberg - Rosanna	88.03
Hawthorn East	88.02
Ivanhoe East - Eaglemont	87.02
Malvern East	86.71
Prahran - Windsor	86.69



TOP 10 LIVEABLE & Affordable Suburbs

sa2 affordable & livable score

Heidelberg
Rosanna

Ivanhoe East
Eaglemont

East MELBOURNE

South Yarra

West

Albert Park

Prehran
Windsor

Toorak

Hawthorn East

Malvern
Glen Iris

Malvern East

Malvern - Gle...
88.79325

Albert Park
91.02518

Heidelberg...
88.03429

Malvern East
86.71130

East Melbourne
100.000...

Toorak
99.77594

Hawthorn East
88.01927

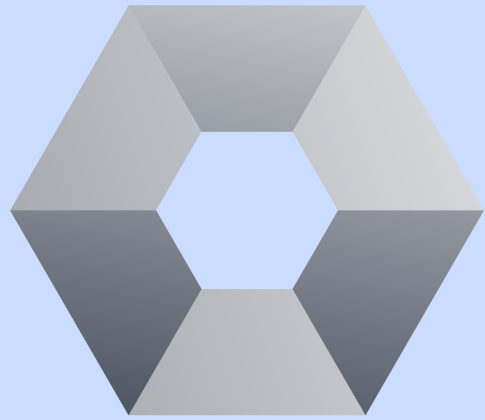
South Yarra...
94.18217

Ivanhoe East...
87.02098

RECOMMENDATIONS



1st - Recommendation



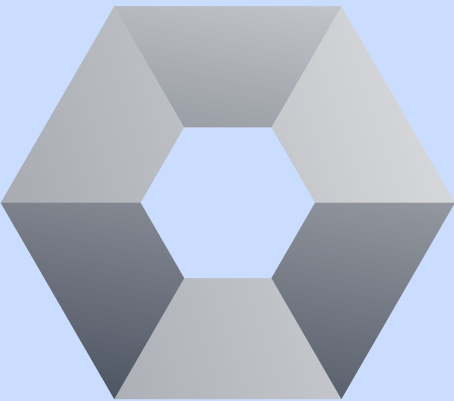
For real estate companies / housing agents / house owners:

Here are **4 important factors** to consider if you want to rent out your property at the **highest possible price**:

- Number of bedroom
- Number of bathroom
- Population density of SA2
- personal income of the SA2

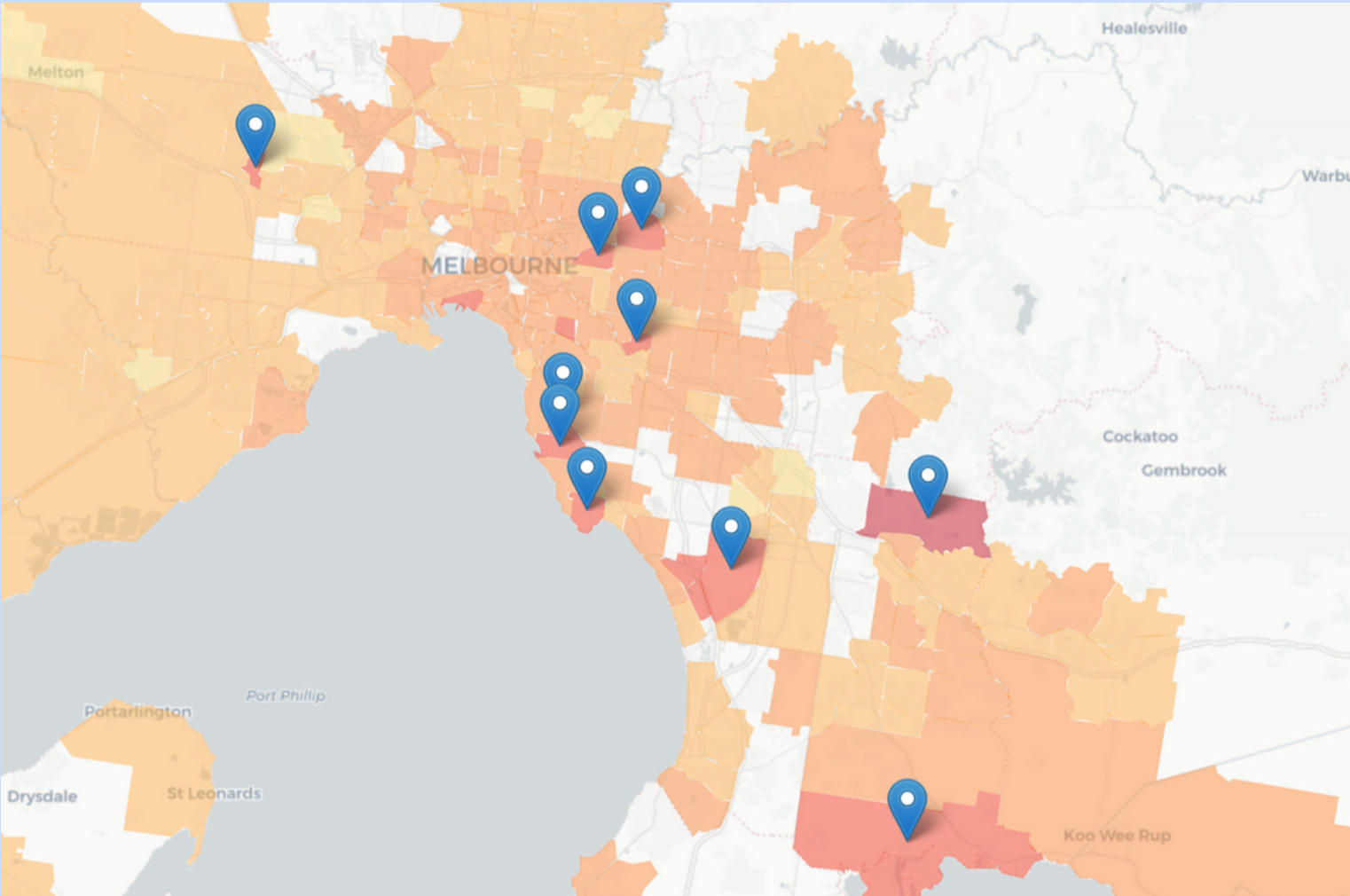
The higher these 4 factors are, the higher price will be.

2nd - Recommendation



For real estate companies / housing agents / house owners:

To rent out the property at the highest possible price, **please consider these 10 suburbs (sa2 districts):**



SA2 Name	Average Price(\$)
Narre Warren North	950
Beaumaris	790
Brighton East	772.50
Ashburton (Vic.)	744.38
Hampton	730.38
Pearcedale - Tooradin	728
Kew - South	718.24
Keysborough - South	709.93
Balwyn North	704
Burnside	700

3rd - Recommendation



For real estate companies / housing agents / house owners:

Consider these 10 suburbs if you want rental prices to **rise as much as possible** over the next 3 years:



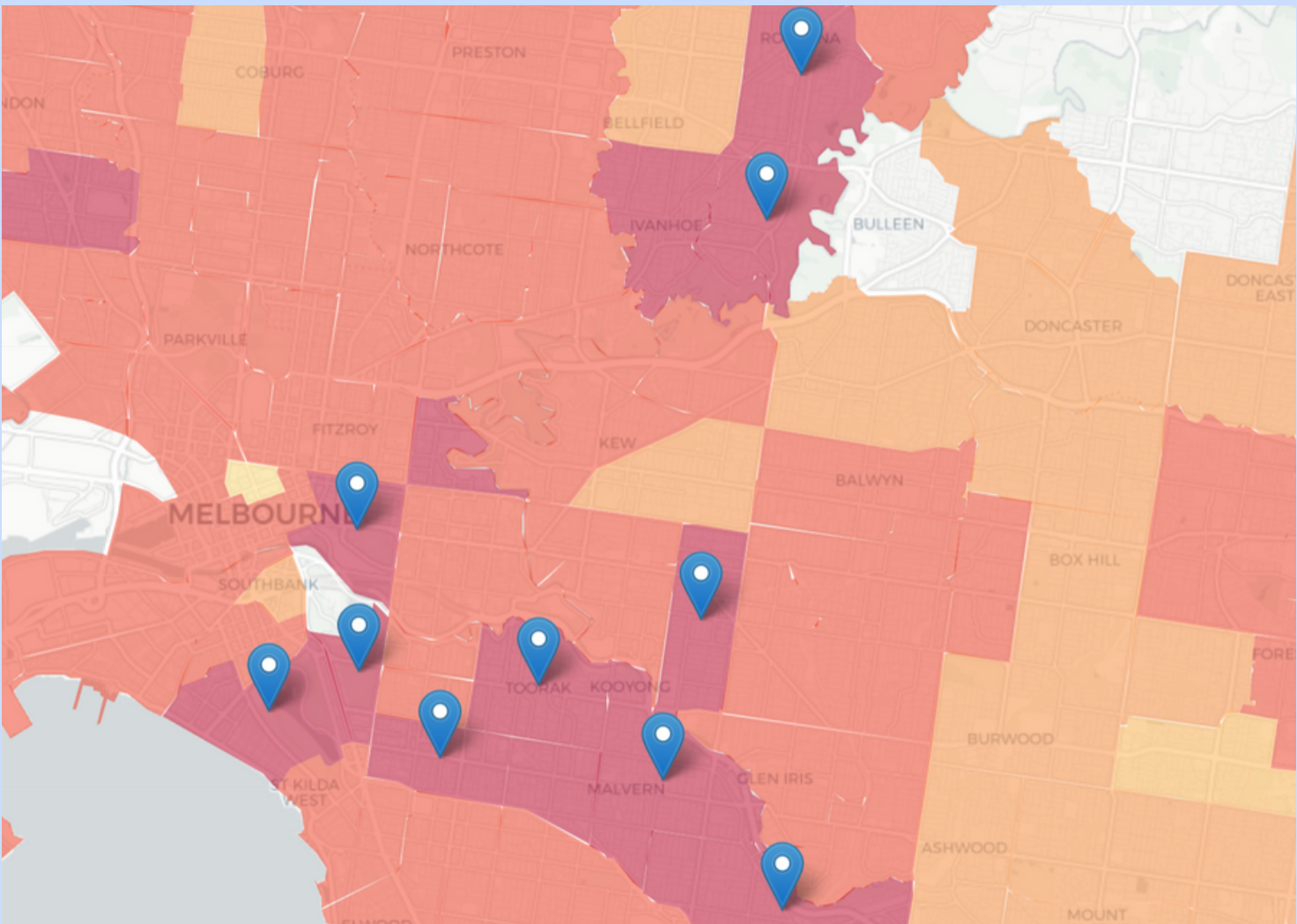
SA2 Name	3yrs Growth Rate(%)
Myrtleford	242.94
Doreen - North	42.45
Doreen - South	37.31
Point Cook - South	34.57
Moe - Newborough	30.49
Seabrook	26.92
Reservoir - North West	26.09
Bundoora - North	25.99
Hillside	24.43
Heidelberg - Rosanna	24.19

4th - Recommendation



For tenants:

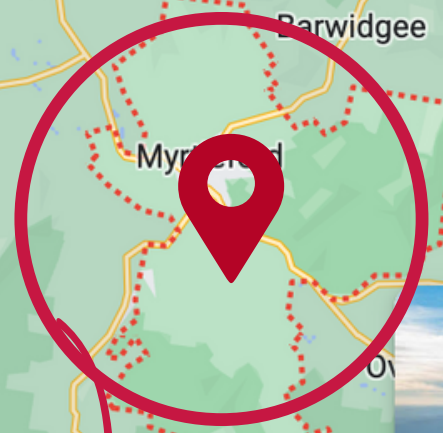
If you want to live in the **most liveable & affordable suburb**, these 10 suburbs are the best:



SA2 Name	Affordable & Liveable Score
East Melbourne	100
Toorak	99.78
South Yarra - West	94.18
Albert Park	91.03
Malvern - Glen Iris	88.79
Heidelberg - Rosanna	88.03
Hawthorn East	88.02
Ivanhoe East - Eaglemont	87.02
Malvern East	86.71
Prahran - Windsor	86.69

Additional Insight

Myrtleford
3-Year Growth Rate
242.9%





Mount Buffalo National Park

4.8 ★★★★★ (1,110)

National park

Open 24 hours

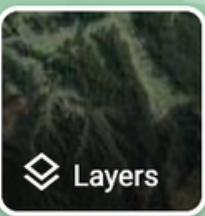


Alpine National Park

4.8 ★★★★★ (1,370)

National park

Open 24 hours



3 Reasons:



Minor Recommendations:

- **For real estate companies:**
Please consider the properties in this suburb and other suburb around this national park.
- **For outdoor enthusiasts:**
Please bring your tent instead of renting an apartment.

REFLECTIONS



SECURITY AND PRIVACY



Access Control:

Restrict access to data and code to authorised personnel with a valid requirement.



Protection of API keys:

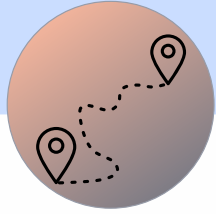
API keys should be kept confidential and not exposed in code repositories or shared openly.



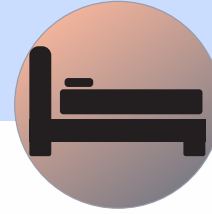
Data Repository Security:

All data is stored on a private Github repository

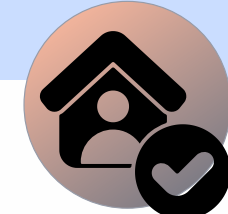
Major Assumption



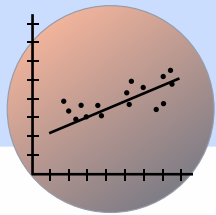
The distance from the property to some facilities **will not change.**



An apartment with **at least 1 bedroom** and **1 bathroom** is livable. (We don't focus on the properties which are unlivable)



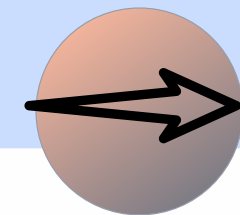
The metric to find the most livable & affordable suburbs we design, is from the **perspective of the tenant**, rather than the real estate companies.



The changes of population density, personal income and offence count **are linear.**



The **offence count** in a postcode is **distributed equally** to each sa2 area it contains.



Compared to walking distance, straight distance **can also reflect the trends.**

Limitation on Ability and Knowledge

- **Limited useful internal features** such as ``if_furnished`` & ``if_balcony`` lead to prediction not accurate enough.
- **Limited to add more useful external features** by using datasets such as ``sa2_expenditures`` & ``sa2_crime_rate`` lead to partial information not be explained
- **Limited to create the 2026 data using a time series model** (e.g. HMM) lead to linear rates of change



DIFFICULTIES



Understaffed

ONE of us suddenly dropped this subject in **WEEK 6**, it led to a disruption in our progress and management.

Everyone spent much more time on the project.



Web Crawling

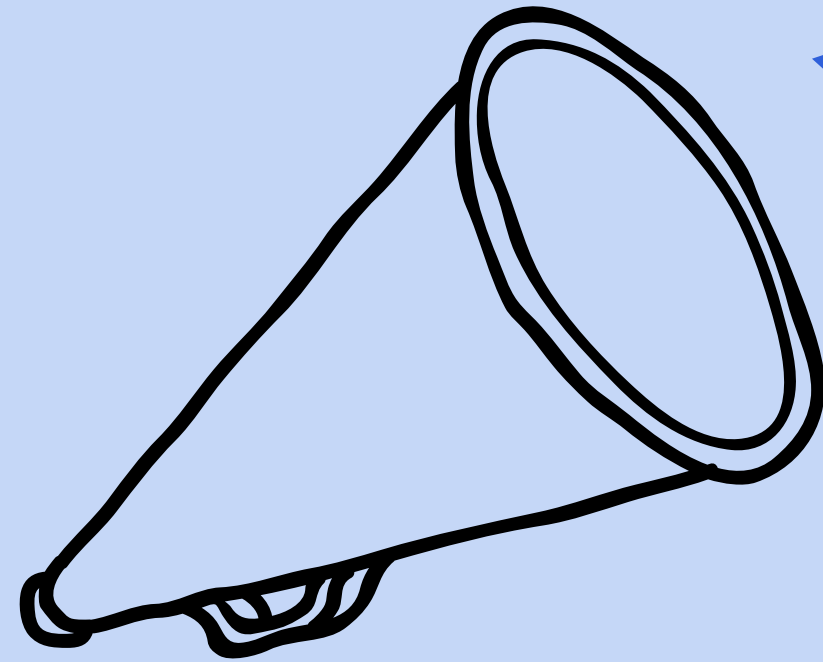
Much useful info is hard to be extracted.



Website Access Restrictions

OpenRouteService website (API) cannot be conveniently access due to the restriction.

WORTH PURSUING???



YES!!!

Even though there exists the limitations & difficulties above, our project has relatively accurate prediction & valuable analysis.

With more specific & targeted analysis, we can give much more recommendations for different groups of people.



Group 22

06-10-2023



**THANKS FOR
WATCHING**

Reference:

- HeiGIT gGmbH. (n.d.). Calculating distance between properties and station. Openrouteservice. <https://openrouteservice.org/>
- Australia-Shoppings.com. List of Victoria (Australia) shopping centres, malls - mall list, locations, hours | Australia Shoppings. (n.d.). <https://www.australia-shoppings.com/malls-centres/victoria>
- Australia's national statistical agency. (n.d.). Geometry shape file. Australian Bureau of Statistics. <https://www.abs.gov.au/>